1. Basic concepts

Term	Definition in English
Embedded system	A computer system designed to perform a specific task inside a device.
Microcontroller (MCU)	A small computer on a single chip, used to control electronic devices.
Microprocessor	A processing chip that executes instructions and usually works with external memory and peripherals.
Single Board Computer (SBC)	A complete computer built on a single circuit board.
Development board	An electronic board used to test, develop and prototype applications.
Firmware	Software stored in an electronic device to control how it works.
Real-time system	A system that must respond to events within a specific time limit.
Bare-metal programming	Programming a device without using an operating system.

2. Hardware vocabulary

Term	Definition in English
CPU	The main processing unit that executes instructions.
Clock frequency	The number of clock cycles per second, measured in hertz (Hz).
RAM	Temporary memory used to store data while a program is running.
Flash memory	Non-volatile memory used to store programs and data.
GPIO	General-purpose pins used to read inputs and control outputs.
ADC	An electronic circuit that converts an analog signal into a digital value.
DAC	An electronic circuit that converts a digital value into an analog signal.
PWM	A digital signal used to control the average power delivered to a device.
Power supply	A device that provides electrical power to a system.
Power consumption	The amount of electrical power used by a device.

3. Communication interfaces

Term	Definition in English
UART	A serial communication interface used to send and receive data between devices.
SPI	A fast serial communication interface used to exchange data between electronic components.
I2C	A serial communication bus that connects several devices using two main wires.

Term	Definition in English
USB	A standard interface used to connect devices and transfer data.
Ethernet	A wired technology used to connect devices to a local network.
Wi-Fi	A wireless technology used to connect devices to a network.
Bluetooth	A wireless technology used for short-range communication between devices.
CAN bus	A communication network widely used in vehicles and industrial systems.

4. Software and development

Term	Definition in English
IDE	Software used to write, edit, compile and debug programs.
Compiler	A program that translates source code into another form of code, often machine code.
Library	A collection of reusable code that provides ready-made functions.
Open-source	Software whose source code is available under a licence allowing its use and modification.
Debugging	The process of finding and fixing errors in a program.
Upload code	To transfer a program from a computer to an electronic board.
SDK	A set of tools and libraries used to develop applications for a specific platform.

5. Applications

Term	Definition in English
Home automation	The use of technology to automatically control devices in a house.
Robot controller	An electronic system that controls a robot's movements and actions.
Data acquisition	The process of collecting data from sensors or other sources.
IoT device	An electronic device connected to the Internet that can send or receive data.
Industrial control	The use of electronic systems to monitor and control industrial processes.
Prototype	An early version of a product used for testing and evaluation.